

RAHMANI MOHAMED REDA || SOFTWARE ENGINEER

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SUMMARY

Solana Smart Contract & SVM Engineer (Rust) working close to validator/RPC systems. I build **Anchor/Pinocchio** programs and **real-time DeFi indexing/monitoring pipelines** for token/DEX activity, liquidity pools, lending/perps, and MEV patterns—grounded in strong **Solana runtime understanding** and **performance optimization** (CU, latency, reliability). I'm an active **open-source contributor**, ship security-first code, and I'm always pushing for **innovation** in the Solana ecosystem.

SKILLS

- **Languages:** Rust (primary) · TypeScript/JavaScript · Python · C/C++
- **On-chain (Solana):** Anchor · Pinocchio · PDAs · CPI · SPL Token · Borsh/zero-copy · program deploy/upgrade · compute optimization · invariants/audits
- **Testing & Tooling:** Mollusk · LiteSVM · Surfpool
- **DeFi:** AMMs/liquidity pools · vault/staking · escrow · lending/borrowing · MEV/Jito bundles
- **Indexing & Pipelines:** transaction decoding · token/DEX indexing · stream processing · backfills/DLQ patterns
- **Data & Messaging:** Kafka · ClickHouse · NATS · gRPC
- **App & Integrations:** Solana web3.js · wallets (Phantom/Solflare) · Next.js/React · Solana Pay · Metaplex
- **Privacy:** Light Protocol zk-compression · Arcium MPC
- **Storage & Databases:** PostgreSQL · MongoDB · AstraDB · Pinecone · Walrus (blob storage)
- **Networking:** QUIC / HTTP3 · Quinn (Rust) · TLS · UDP/TCP
- **DevOps/Cloud:** Docker · Git/GitHub · CI/CD · Jenkins · Azure · AWS (IAM, VPC, EC2, S3, CloudWatch, ECR, RDS)

EXPERIENCE

Backend Engineer (GenAI Platform) – Niloom AI (Remote, New York) | Jun 2024 – Present

- Built and maintained backend services integrating multiple LLM providers (OpenAI, Mistral, LLaMA) using **LangChain/LangGraph**.
- Implemented **embeddings + similarity search** to improve retrieval quality and response relevance.
- Designed **RAG pipelines** and orchestration workflows (prompting + chaining) for reliable task execution.
- Added automated evaluation and regression tests (**RAGAS**) to track answer quality and prevent performance drift.
- Delivered features in an **Agile** workflow with strong focus on reliability, testing, and maintainable code.
- Developed data/storage layers using **PostgreSQL** and **MongoDB**, and managed artifacts/documents via **AWS S3**.
- Containerized services with **Docker** and supported **CI/CD** pipelines for reliable deployments and fast iteration.

MERN Stack Developer Intern — Ark X | Jan 2024 – Jun 2024

- Built and shipped full-stack features using **MongoDB, Express.js, React, and Node.js** (MERN).
- Designed and maintained **MongoDB schemas** and indexes to improve query efficiency.
- Implemented REST APIs and backend business logic, including validation and error handling.
- Developed responsive React UI components and integrated them with backend endpoints.
- Collaborated in a team workflow (code reviews, Git, issue tracking) to deliver weekly milestones.

Software Engineer (Bootcamp) — ALX Africa | Feb 2023 – Apr 2024

- Completed an intensive full-stack program covering web development with **JavaScript/TypeScript, React/Next.js, Node.js/Express**, plus backend development in **Python (Django, Flask)**.
- Built and shipped multiple projects involving **API design**, authentication, and database-backed applications using **SQL and NoSQL** data stores.
- Developed systems fundamentals through projects in **C, Unix/Linux**, and **shell scripting (Bash)**, emphasizing debugging, memory/process concepts, and automation.
- Applied **networking** concepts (TCP/IP, HTTP, DNS) and DevOps practices such as **Git, CI workflows, and deployment basics** in team projects.

EDUCATION

- **Rektoff**
- **Turbin3**
- **ALX Africa**
- **JobInTech**
- **Higher School of Technology (EST), Oujda**
- **Solana Rust Security Bootcamp (Ongoing) | Cohort 3 | Present**
- **Builders Cohort + Advanced SVM Cohort | Apr 2025 – Aug 2025**
- **Software Engineering Program | Feb 2023 – Apr 2024**
- **Full-Stack MERN Program | Jan 2024 – Jun 2024**
- **B.Eng. Network Security | Aug 2022 – Jul 2023**

PROJECTS

- **RLock — Intent-aware Solana bundler & optimizer (Rust / TypeScript)** Built an open-source bundler that turns high-level intents into a deterministic execution plan (dependency/conflict DAG + topological scheduling). Chunks multi-step flows into conflict-aware bundles, respecting Solana constraints (CU, tx size, account limits) and using ALTs/ComputeBudget patterns when needed. Integrated an Ephemeral Rollups-first execution path with a safe L1 RPC fallback, plus OCC-style checks to reduce failures under contention ([deep dive article on Medium](#)) ([Github](#)).
- **Solana DEX MEV Indexer — Real-time decoding + analytics pipeline (Rust / Kafka / ClickHouse):** Built a Rust-based streaming indexer that ingests Solana transactions (RPC / streaming sources) and decodes DEX activity into normalized events. Produces structured datasets for token/DEX indexing (swaps, pools/liquidity movements) and MEV analysis designed for monitoring and research workflows. Implemented a production-style pipeline Kafka → ClickHouse with reliability patterns (schema versioning, backfills, and DLQ-style handling). Currently extending the decoder to extract MEV patterns across major Solana venues (Raydium, Jupiter, Orca) — starting with swap-level signals and expanding toward sandwich/arbitrage-style detection. ([Github](#))